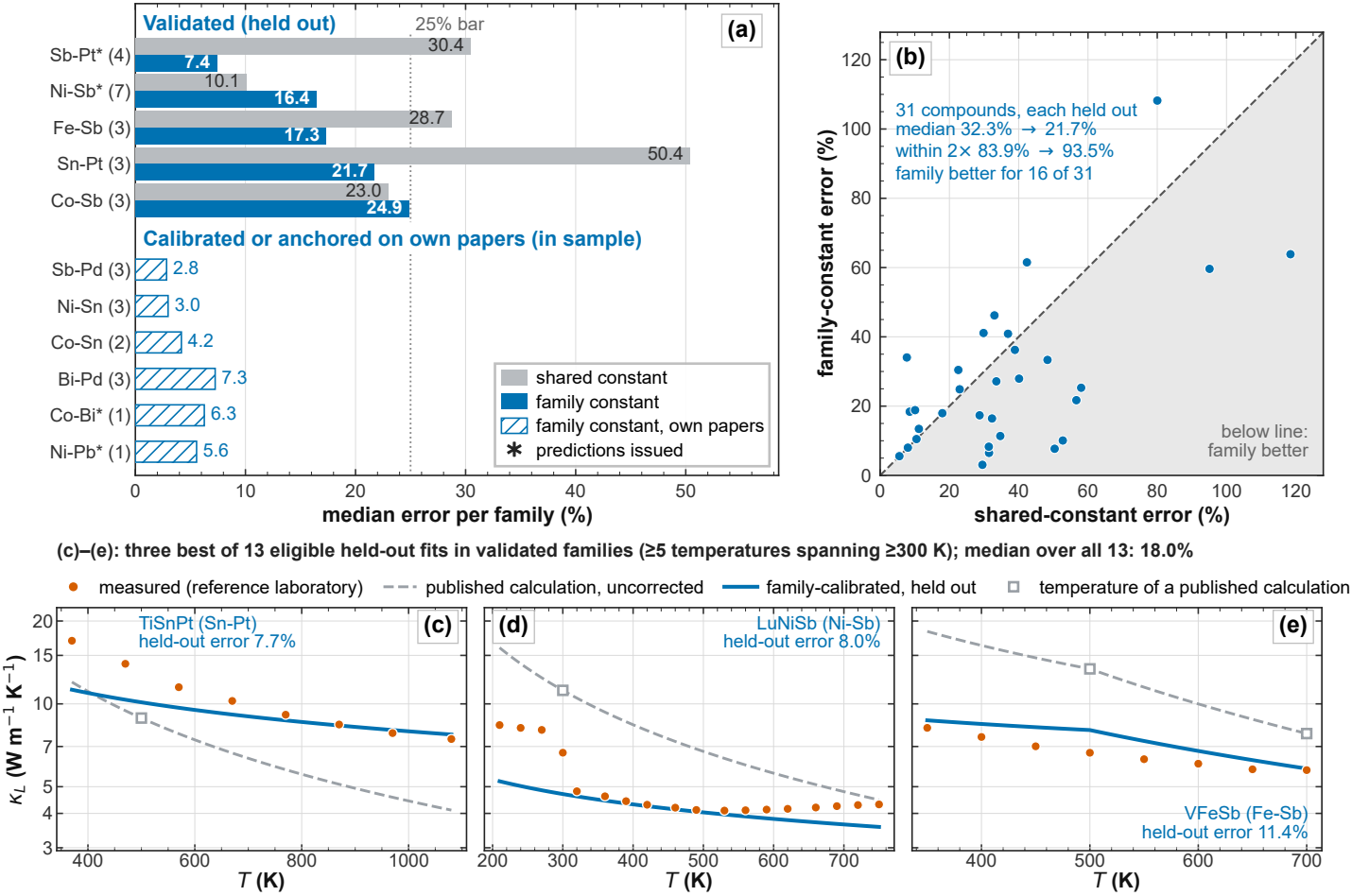

Validated (held out)
25% bar
(a)
Sb-Pt* (4)
7.4
30.4
Ni-Sb* (7)
10.1
16.4
Fe-Sb (3)
28.7
17.3
Sn-Pt (3)
50.4
21.7
Co-Sb (3)
23.0
24.9
Calibrated or anchored on own papers (in sample)
Sb-Pd (3)
2.8
Ni-Sn (3)
3.0
Co-Sn (2)
4.2
Bi-Pd (3)
7.3
Co-Bi* (1)
6.3
Ni-Pb* (1)
5.6
shared constant
family constant
family constant, own papers
predictions issued
0
10
20
30
40
50
median error per family (%)
(b)
31 compounds, each held out
median 32.3% → 21.7%
within 2× 83.9% → 93.5%
family better for 16 of 31
below line:
family better
family-constant error (%)
shared-constant error (%)
0
20
40
60
80
100
120
(c)–(e): three best of 13 eligible held-out fits in validated families (≥5 temperatures spanning ≥300 K); median over all 13: 18.0%
measured (reference laboratory)
published calculation, uncorrected
family-calibrated, held out
temperature of a published calculation
(c)
TiSnPt (Sn-Pt)
held-out error 7.7%
(d)
LuNiSb (Ni-Sb)
held-out error 8.0%
(e)
VFeSb (Fe-Sb)
held-out error 11.4%
κL (W m−1 K−1)
T (K)
3
4
5
7
10
15
20
200
300
400
500
600
700
800
1000